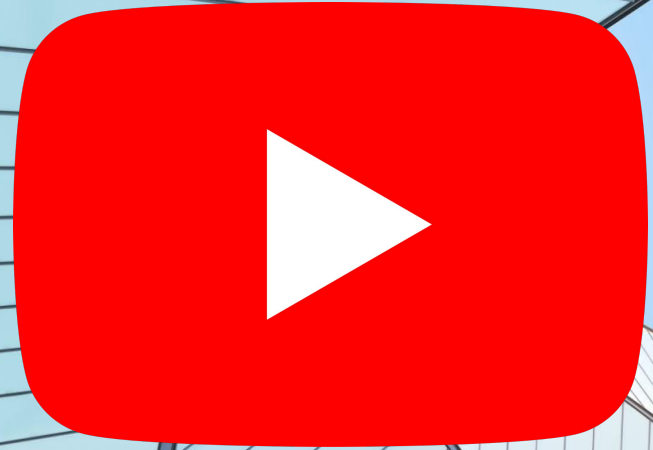


Youtube Channel Data Analytics

Tools Used: Python, Pandas, Matplotlib,
YouTube Data API

By Himanshu Tripathi



Content *overview*

Objective →

Problem Statement →

Metrics / KPIs →

Visualization →

Insights →

Recommandations →

Objectives

- To extract, analyze, and visualize YouTube video/channel data using YouTube Data API
- Help stakeholders understand channel performance
- Identify content trends that drive growth
- Optimize future video strategy based on data insights

Problem Statement

What Are We Solving?

- **No clear visibility into which videos are performing well**
- **Engagement patterns are unclear**
- **Difficult to track growth KPIs over time**
- **Data is spread across multiple videos without consolidated view**

Metrics Tracked

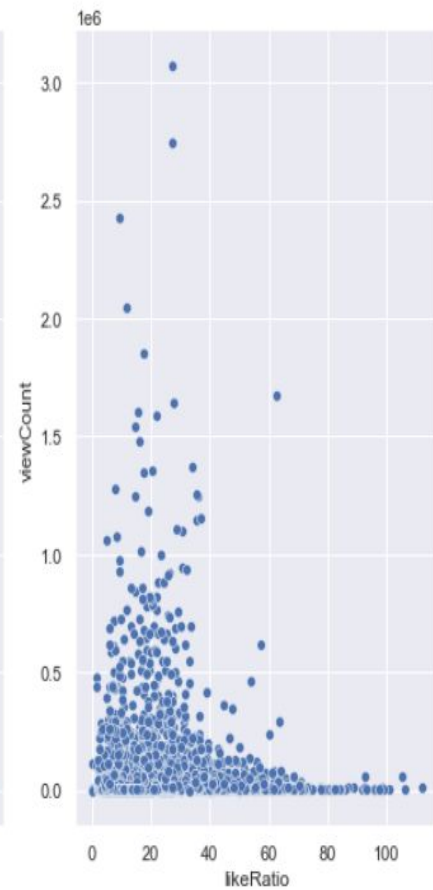
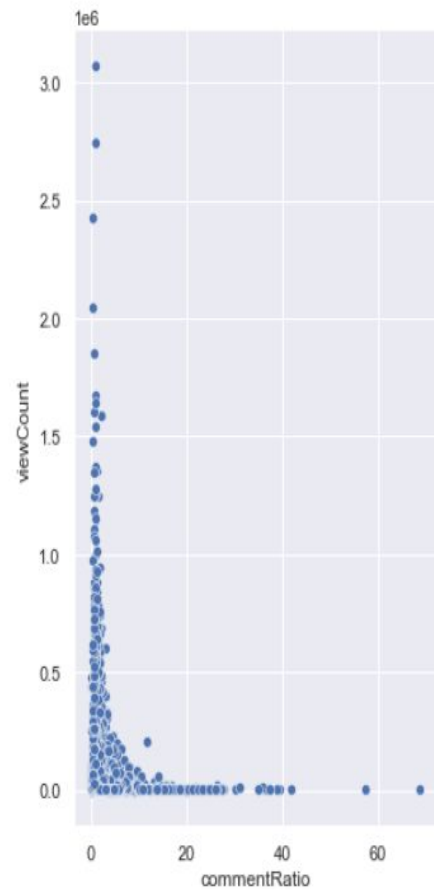
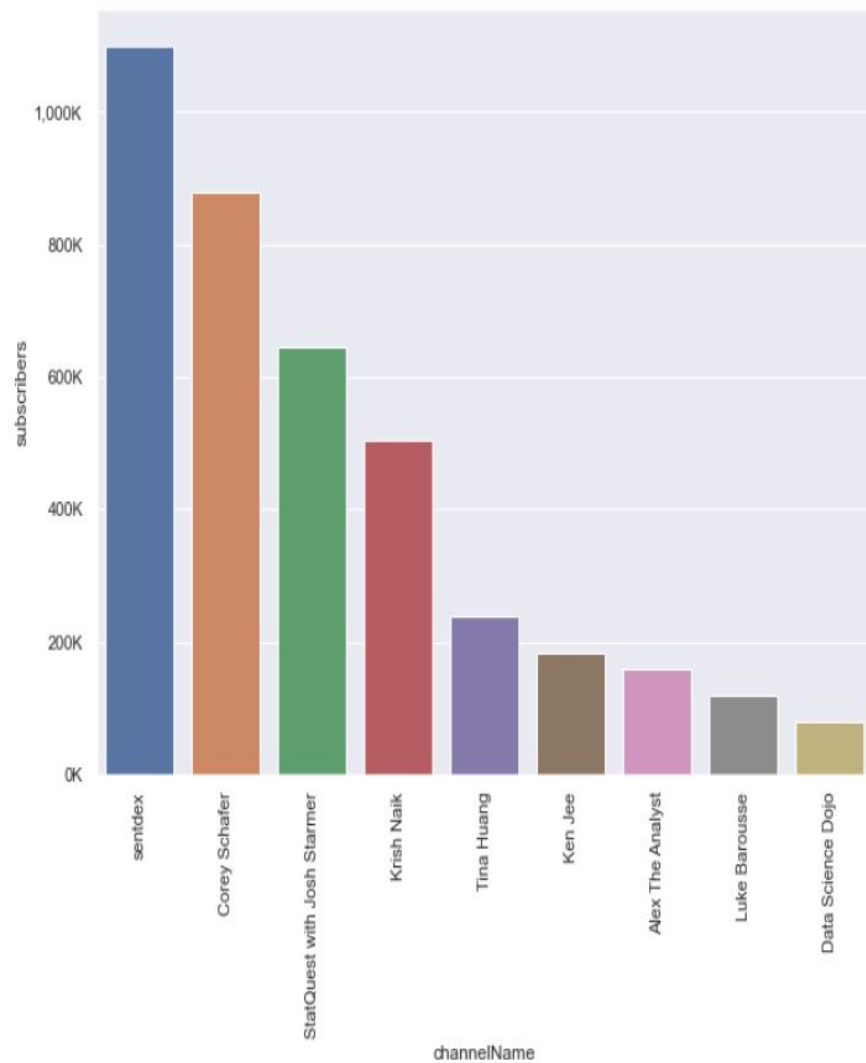
- Video title & publish date
- Views, Likes, Comments
- Duration & Tags
- Video popularity and ranking
- Channel stats: Total views, subscribers, total videos

Key KPIs

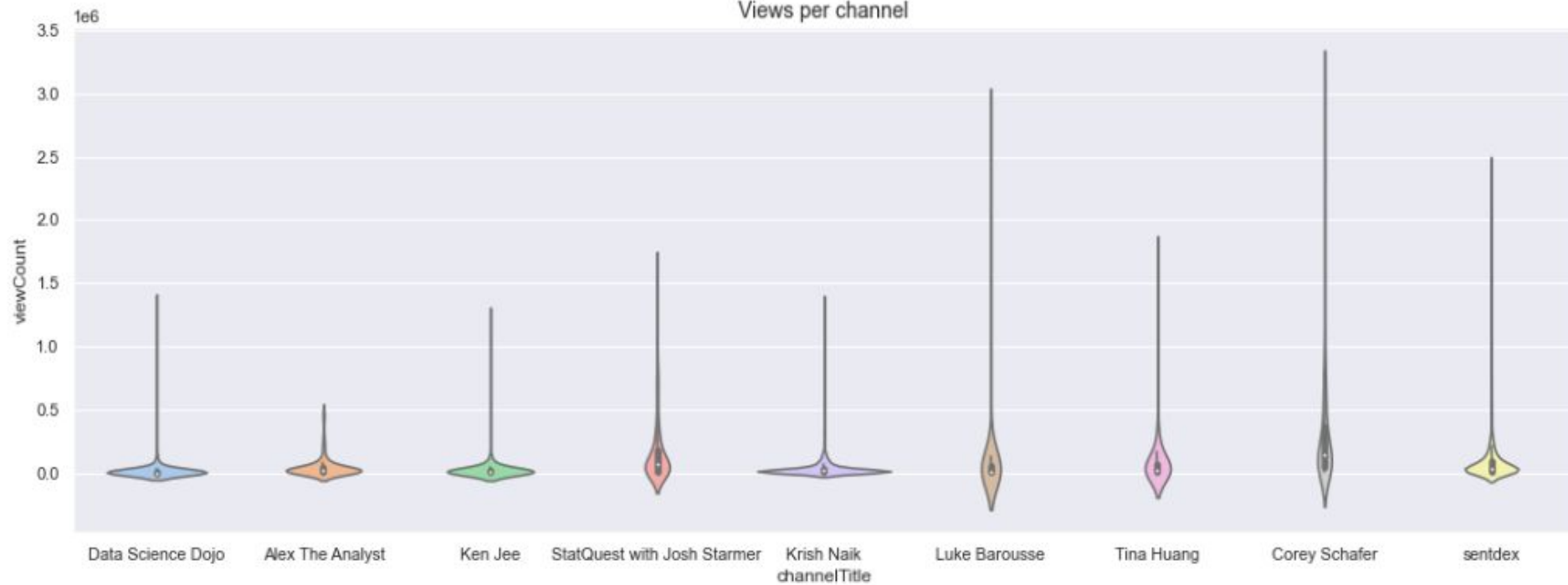
- Total Videos Published
- Average Views per Video
- Most Watched Video
- Engagement Rate = $(\text{Likes} + \text{Comments}) / \text{Views}$
- Publishing Consistency (videos/month)

Graphs & Visualizations

	channelName	subscribers	views	totalVideos	playlistId
0	Alex The Analyst	158000	5900052	126	UU7cs8q-gJRIGwj4A8OmCmXg
1	Luke Barousse	120000	5732718	68	UULLw7jmFsvfIVaUFsLs8mlQ
2	Data Science Dojo	79600	4504751	279	UUzL_0nle8B4-7ShhVPfJkgw
3	StatQuest with Josh Starmer	645000	32254787	211	UUtYLUTtgS3k1Fg4y5tAhLbw
4	sentdex	1100000	99602241	1237	UUfzlCWGWYyIQ0aLC5w48gBQ
5	Krish Naik	504000	41660941	1289	UUNU_IfiiWBdtULKOW6X0Dig
6	Tina Huang	238000	8229073	81	UU2UXDak6o7rBm23k3Vv5dww
7	Corey Schafer	879000	67492239	230	UUCezlgC97PvUuR4_gbFUs5g
8	Ken Jee	182000	5532488	221	UUiT9RITQ9PW6BhXK0y2jaeg



Views per channel



References/ Resources used:

[1] Youtube API. Available at <https://developers.google.com/youtube/v3>

[2] Converting video durations to time function.

<https://stackoverflow.com/questions/15596753/how-do-i-get-video-durations-with-youtube-api-version-3>

[3] P. Covington, J. Adams, E. Sargin. The youtube video recommendation system. In Proceedings of the Fourth ACM Conference on Recommender Systems, RecSys '16, pages 191-198, New York, NY, USA, 2016. ACM.

Key Insights

What data tells us

- 80% of views come from 20% of videos
- Videos under 10 minutes have higher engagement
- Posting mid-week (Wed–Thu) = more views
- Tags like “Python”, “API”, and “Tutorial” are most used in top videos

Recommendations

- 1 Focus on short, high-value content
- 2 Increase publishing on peak days (Tue–Thu)
- 3 Optimize video titles + descriptions with SEO keywords
- 4 Promote top-performing videos across other platforms
- 5 Monitor engagement-to-view ratio for future decisions

Thank you